

3(a)	any two from: substrate-linked (phosphorylation) ; A substrate level phosphorylation chemiosmosis / oxidative (phosphorylation) ;	2
3(b)	any six from: 1 hydrogen / electron, carriers ; R hydrogen ions / hydrogen molecules 2 in glycolysis NAD becomes reduced ; 3 (so that) triose phosphate becomes, oxidised / dehydrogenated ; 4 in the link reaction NAD becomes reduced ; 5 (so that) pyruvate becomes, oxidised / dehydrogenated or for production of acetyl coenzyme A ; 6 in the Krebs cycle both NAD and FAD become reduced ; 7 to regenerate oxaloacetate ; 8 (deliver, hydrogen / H^+ and e^-), to inner mitochondrial membrane / to cristae / to ETC / for oxidative phosphorylation / for chemiosmosis ; 9 <i>ref. to</i> ATP production ; 10 <i>ref. to</i> recycling of, NAD / FAD ;	6

Question	Answer	Marks
3(c)	<i>any two from:</i> 1 <i>ref. to increase / decrease / control, of (rate of the) link reaction ;</i> 2 <i>allows build-up of acetyl CoA to be used in the Krebs cycle ;</i> 3 <i>enzyme becomes active again when, coenzyme A increases / ratio falls ;</i> 4 <i>allows more coenzyme A, to enter / return to, the link reaction ;</i> <i>A not enough CoA to enter the link reaction ora</i> 5 <i>AVP ; e.g. end product inhibition</i>	2